

Bitcoin, Cryptocurrencies, and Blockchain technology

1. Bitcoin & Cryptocurrencies
2. Blockchain Technology
3. Security and Privacy in Bitcoin and Cryptocurrencies

Lecture 5 - Review

Main points to take home

▶ A big picture of Public key Cryptography

- ▶ A party owns a key pair (Public key, Private key)
- ▶ Encrypt using Public key, decrypt using Private key

▶ Diffie Hellman Key Exchange protocol

- Public channel can be used to create a confidential shared key function.
- One-way operation should be used
 - Mix colour example
- In practice: we use multiplicative group of integers modulo p , where p is prime, and g is a primitive root modulo p

Main points to take home

► Diffie Hellman Key Exchange protocol

- Multiplicative group of “Integers modulo p ”

$$s = g^n \bmod p$$

- We have:

Easy: given g , n , & p , solve for s

Hard: given s , g , & p , solve for n

- And the property of:

$$g^{a*b} \bmod p = g^{b*a} \bmod p$$

RSA Summary

- ▶ RSA algorithm

- ▶ How to generate the key pair

Public key: $pk = (e, n)$; Private key: $sk = (d, n)$

- ▶ How to encrypt using the public key

Compute ciphertext $C = M^e \bmod n$

- ▶ How to decrypt using the private key

Compute $M' = C^d \bmod n$

$$= (M^e \bmod n)^d \bmod n$$

$$= M^{ed} \bmod n = M$$

Main points to take home

▶ Digital Signature based on RSA:

A triple (Gen, Sign, Ver)

- ▶ Gen(len) : generate a key-pair (pk, sk)
- ▶ Sign(m; sk) : sign a message m with private key sk, output signature s
- ▶ Ver(m, s; pk) : verify signature s with public key pk

Main points to take home

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Main points to take home

➤ Elliptic Curves Cryptography

- Basic Elliptic Curves

$$y^2 = x^3 + Ax + B$$
$$(4A^3 + 27B^2 \neq 0) \text{ (Weierstrass Form)}$$

- Elliptic Curve for Cryptography = Elliptic Curve over a Finite Field L : $E(L)$

$$E(L) = \{\infty\} \cup \{(x, y) \in L \times L \mid y^2 + \dots = x^3 + \dots\}$$

- Combine with “*Integer mod p*”, P is a prime number to form a Finite Field

Main points to take home

➤ Elliptic Curves Cryptography

- Addition and Multiplication on Elliptic Curve
 - How we define $P + Q$ over $E(L)$
 - How we define $mP = P + P + \dots + P$ over $E(L)$
- Elliptic Curves Discrete Logarithm Problem (ECDLP)
- Elliptic Curves Diffie Hellman (ECDH)

Main points to take home

➤ Elliptic Curves Discrete Logarithm Problem (ECDLP)

- There is efficient way to calculate the multiplication on EC:

$$Q = mP, \text{ given } m \text{ and } P$$

- The ECDLP is to find the value of m , given Q and P that

$$Q = mP$$

- Basically, we do not know of a fast way to solve ECDLP.
- This will be good for the trapdoor function (with modular p)
 - m will be the secret information, Q and P and the modulus prime p can be publicly available

Main points to take home

➤ **Elliptic Curves Diffie Hellman Key Exchange**

- Alice and Bob desire to establish a shared secret key k .
- Alice and Bob agree on an EC, a large prime p (called the multiplicative order), and a point B on the curve (called the Generator or the Base-point)
 - Called Domain Parameters
- Eve knows the curve, the point, and the prime number

Main points to take home

- Alice secretly picks a large number p_A . Bob secretly picks a large number p_B . Alice computes $Q_A = p_A B$. Bob computes $Q_B = p_B B$. They exchange the points Q_A and Q_B .
 - p : private key
 - $Q = pB$: public key
- Alice computes $p_A Q_B = p_A p_B B$. Bob computes $p_B Q_A = p_B p_A B$. Both use the x value of $p_A p_B B$ for the key k .
- The shared secret key k is then used for symmetric encryption such as DES and AES

Lecture 6:

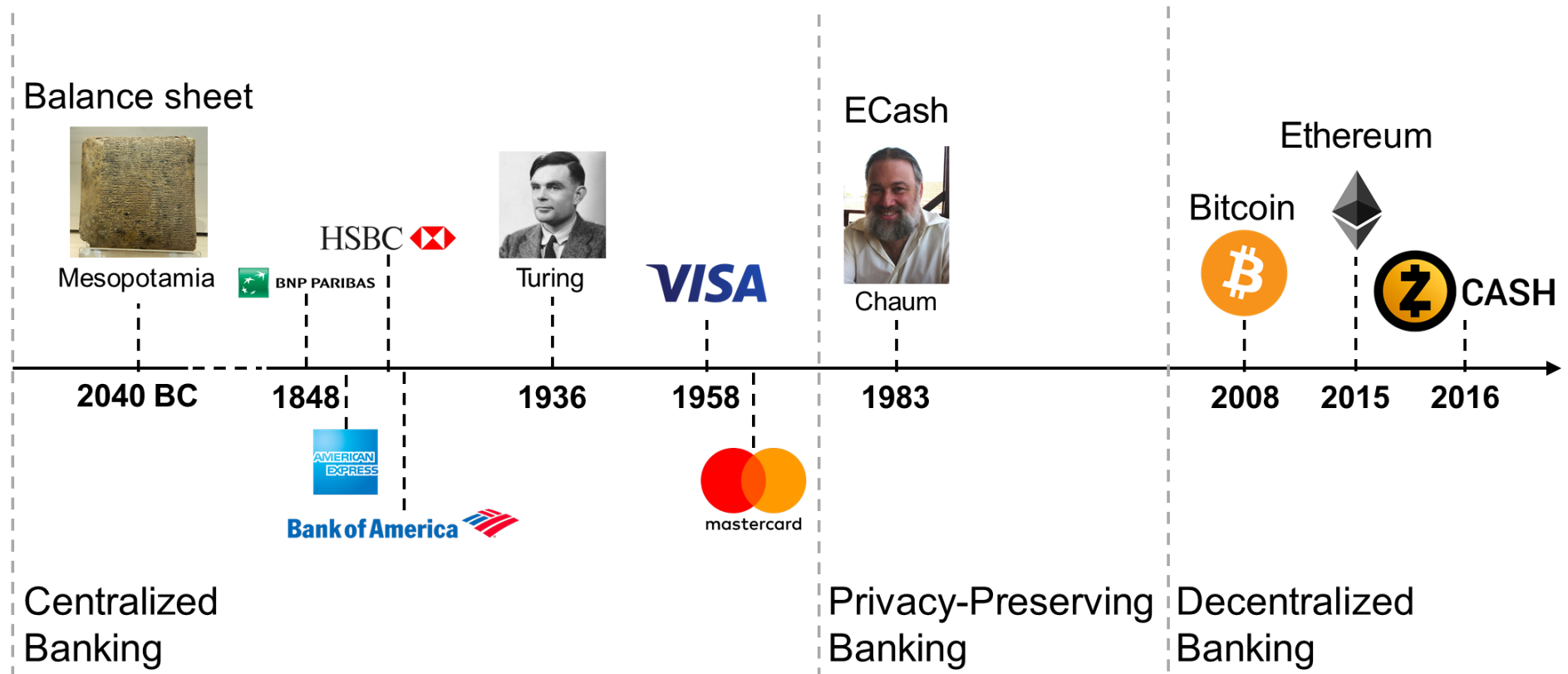
Bitcoin, Cryptocurrencies, and Blockchain technology

Outline

1. Bitcoin & Cryptocurrencies
 - Decentralised Payment System: A Big Picture
 - Cryptography in Bitcoin
 - Bitcoin Protocol: an Overview
2. Blockchain Technology
 - Bitcoin Addresses
 - Bitcoin Distributed Ledger: a Blockchain
 - Transactions in Bitcoin and Blockchain
 - Consensus Mechanism: Proof-of-Work
3. Security and Privacy in Bitcoin and Blockchain

Bitcoin and Cryptocurrencies

History of Banking/Payment System



Money

➤ What is Money

- Coins, Bill – can't exist on two places at one time
- Bearer bonds: immediate cashable
- Gold, Silver: portable money

➤ Features:

- Difficulty to counterfeit
- Immediate settle-able
- Untraceable.

Digital Cash

- Two types of Digital Cash
 - Identified digital cash
 - Anonymous digital cash
- Ideal Digital Cash system:
 - Independency (transfer through network)
 - Security (not copied or re-used)
 - Privacy (untraceable)
 - Offline payment (peer to peer)
 - Transferability (to another person)
 - Divisibility (digital cash can be divided into smaller amount)

Digital Cash

- Digital cash is like other cash.
 - We need to buy some initial digital coins with another form of money before we can spend them.
- Preventing forgery is harder because it is easier to duplicate a sequence of bits.
- Anonymity is harder since anything that happens online can leave a trace.
- The easiest ways of implementing digital cash is to use a trusted central authority.
 - This has little advantage over just using a bank and credit and debit cards.

Protocol Consideration

➤ Consumer:

- Privacy; Security; Protection; Regulation

➤ Business:

- Availability of anonymity; Cost and ease of acquisition; Availability; Risk of fraud; Liability for fraud.

➤ Financial And Government:

- Consumer protection; Financial loss; Privacy vs fighting crime; Federal reserve regarding the money.

➤ Technical Challenges:

- Anonymous spending; Privacy; Preventing fraud, like double spending; Cost effective

David Chaum and DigiCash

- Blind Signature: to provide anonymity
 - What a normal protocol looks like (using PKI)
 1. Alice: sign for a note with money order + SerialNum
 2. Bank: verify and sign the note + deduct her account
 3. Alice use the note signed by bank to Bob for service or product
 4. Bob verify bank signature, and send note to bank
 5. Bank verify its signature and SerialNum not claimed before, put proper money into Bob's account

David Chaum and E-Cash

➤ Change above to provide anonymity

1. Alice: sign for a note with money order + SerialNu. Do this for 100 envelop for instance. Generating Anonymous by process of blinding.
2. Bank: verify the note by read 99 envelop (cut-and choose protocol, ask Alice to unblind those 99) + deduct her account + sign the last one without open
3. Alice use the note (by unblinding it) signed by bank to Bob for service or product.
4. Bob verify bank signature, and send note to bank
5. Bank verify its signature and SerialNum not claimed before, put proper money into Bob's account. However, bank won't know where the original money come from (it doesn't know the SerialNum)

David Chaum and E-Cash

➤ Prevent double-spending

- Bank needs to know who the double spender is: Alice or Bob?
 1. Alice: sign for n notes with money order + SerialNum + Identity info pairs (identity pair using secrete splitting and bit commitment protocol)
 2. Bank: ask Alice to unblind n-1 notes and reveal identity; when satisfy, sign the last blind note.
 3. Alice unblind the note and give to Bob
 4. Bob verify bank signature
 5. Bank verify its signature and SerialNum not claimed before, put proper money into Bob's account.

Detecting the culprit if SerialNum is re-used.

E-Cash Crypto Protocols

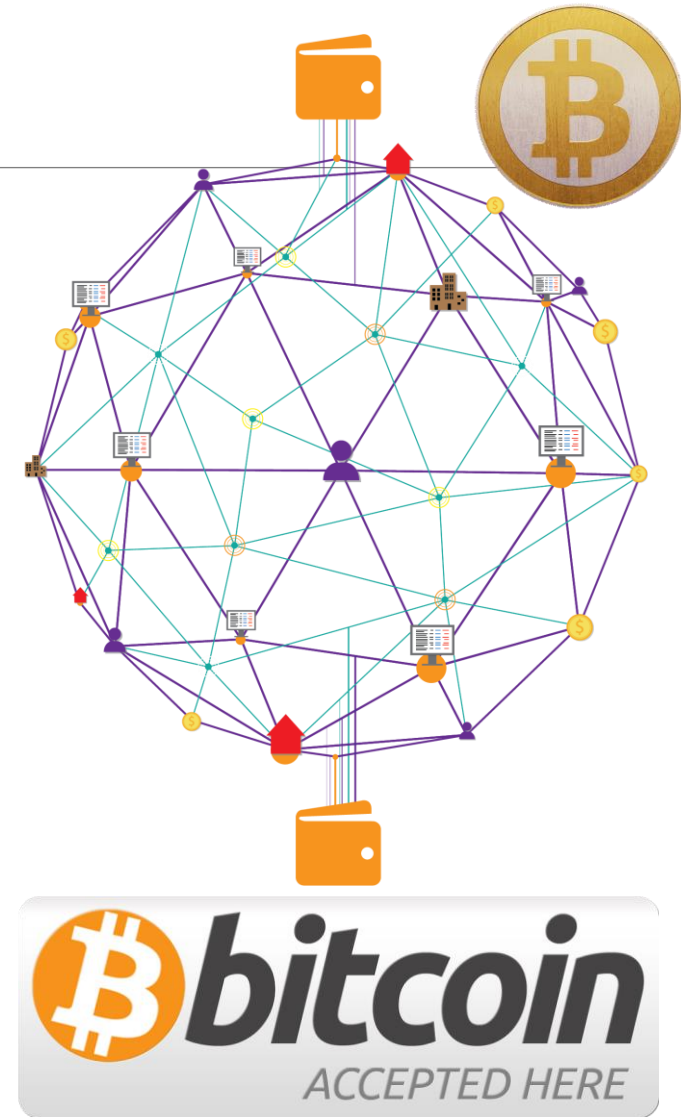
- Various practical issues:
 - Need for trusted central party
 - Computationally expensive
 - Etc.

Ideas of Cryptocurrencies

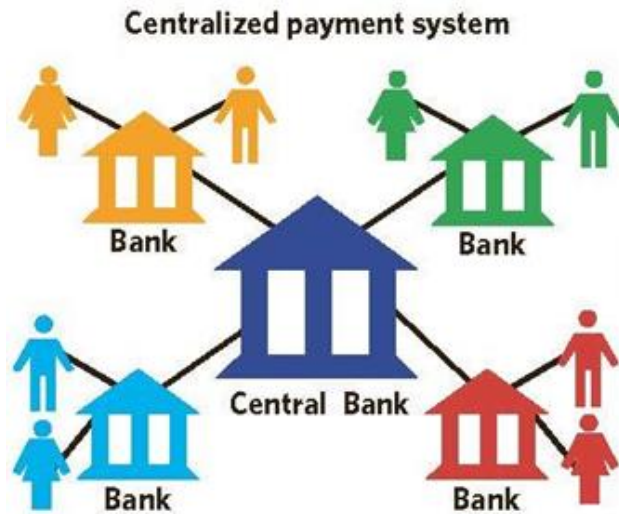
- They don't have a central authority.
 - They use a peer-to-peer system with all peers equal.
- This requires enough of the peers to agree on which transactions have happened.
 - A ledger that can't be altered.
 - How are disagreements resolved?
- They also need techniques to prevent forgery and double spending.
- They should also prevent a denial-of-service attack
 - So that someone is not prevented from spending their money.

What is Bitcoin

- A **distributed, decentralized** cryptocurrency system
- Effectively “**a bank**” run by an “**ad-hoc (peer-to-peer) network**”
 - Digital checks
 - A distributed transaction log (i.e., ledger)
 - Nodes running Bitcoin protocol (lightweight, full nodes)
- Released by **Satoshi Nakamoto** 2008

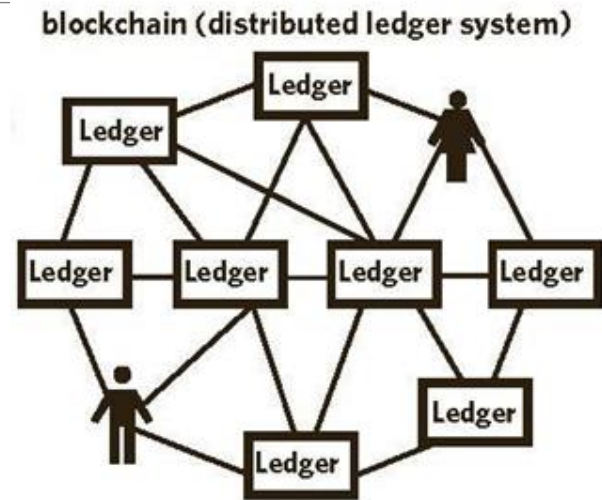


Bitcoin vs Traditional (Centralised) Banking System



Centralised ledger:

- **Identity:** sort-code/account number
- **Authentication:** physical bank card, internet/mobile banking account
- **Transactions:** verified and recorded (on to the ledger) by a commercial & central bank



Distributed Ledger

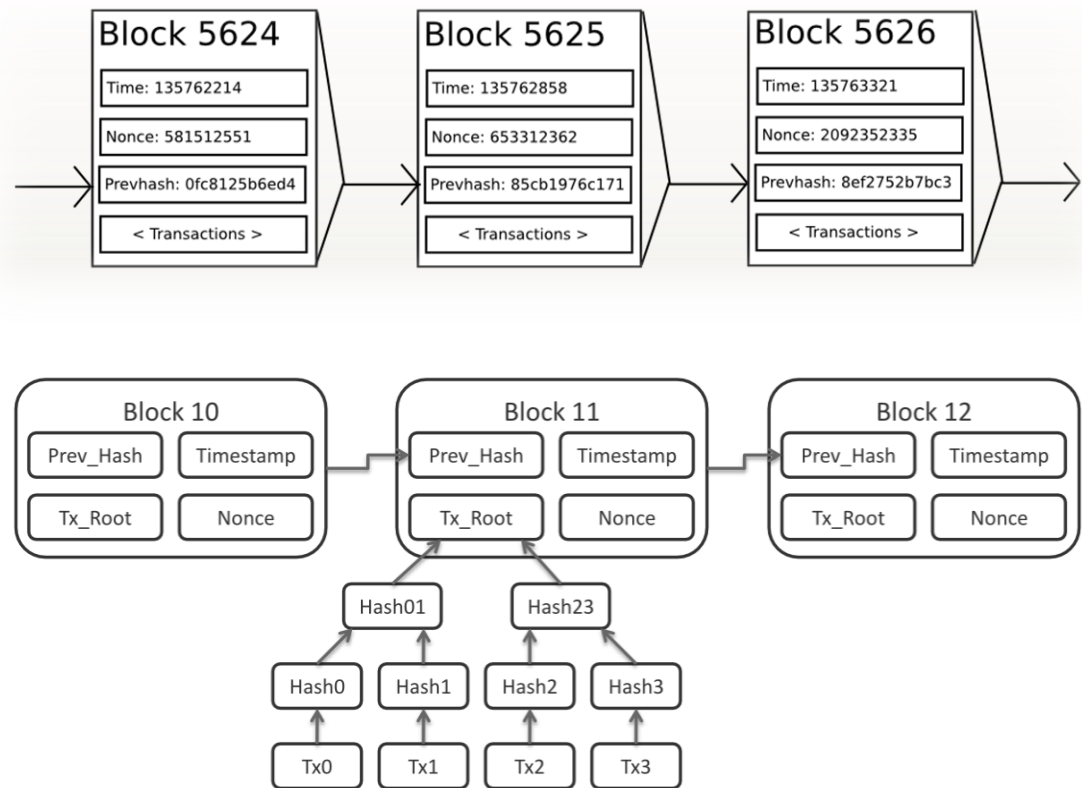
- **Identity:** public-key (e.g., Bitcoin Address = hash of hash of public-key)
- **Authentication:** digital signature
- **Transactions:** verified and recorded (onto distributed ledger (i.e., Bitcoin blockchain) by participants in the network.

Bitcoin vs Traditional (Centralised) Banking System

First National Bank				
1234 First Avenue Primo Vista, CA 90783-1409				
Statement of Account 109-654-5454-45		April 30, 20X8		
Vector Management Group 3214 Tangent Ln. Circle Park, CA 90778-3421		Balance Last Statement	7,358	
		Total Credits	14,083	
		Total Debits	13,239	
		Balance This Statement	8,202	
Date	Check	Debits	Credits	Balance
4/1/20X8				7,358
4/2/20X8			3,200	10,558
4/2/20X8	1541	152		10,406
4/4/20X8	1547	330		10,076
4/5/20X8	1551	18		10,058
4/6/20X8		20 SC		10,038
4/6/20X8	1553	152		9,886
4/7/20X8	1554	87		9,799
4/9/20X8			2,800	12,599
4/10/20X8	1555	1,524		11,075
4/11/20X8	1556	765		10,310
4/12/20X8		253 DM		10,057
4/12/20X8	1557	32		10,025
4/13/20X8	1558	304		9,721
4/14/20X8	1559	3,227		6,494
4/16/20X8			3,100	9,594
4/17/20X8	1560	81		9,513
4/19/20X8		50 SC		9,463
4/19/20X8	1561	152		9,311
4/20/20X8	1562	66		9,245
4/20/20X8	1563	1,325		7,920
4/21/20X8	1566	358		7,562
4/23/20X8			3,400	10,962
4/24/20X8	1567	429		10,533
4/24/20X8		345 NSF		10,188
4/25/20X8			1,565 CM	11,753
4/27/20X8	1568	3,188		8,565
4/30/20X8	1569	381		8,184
4/30/20X8			18 INT	8,202

Symbol key: CM = Credit Memo (see attachment) INT = Interest
DD = Direct Deposit NSF = Not Sufficient Funds
DM = Debit Memo (see attachment) SC = Service Charge

Centralised Ledger



Bitcoin distributed ledger:
chain of block (**Blockchain**)

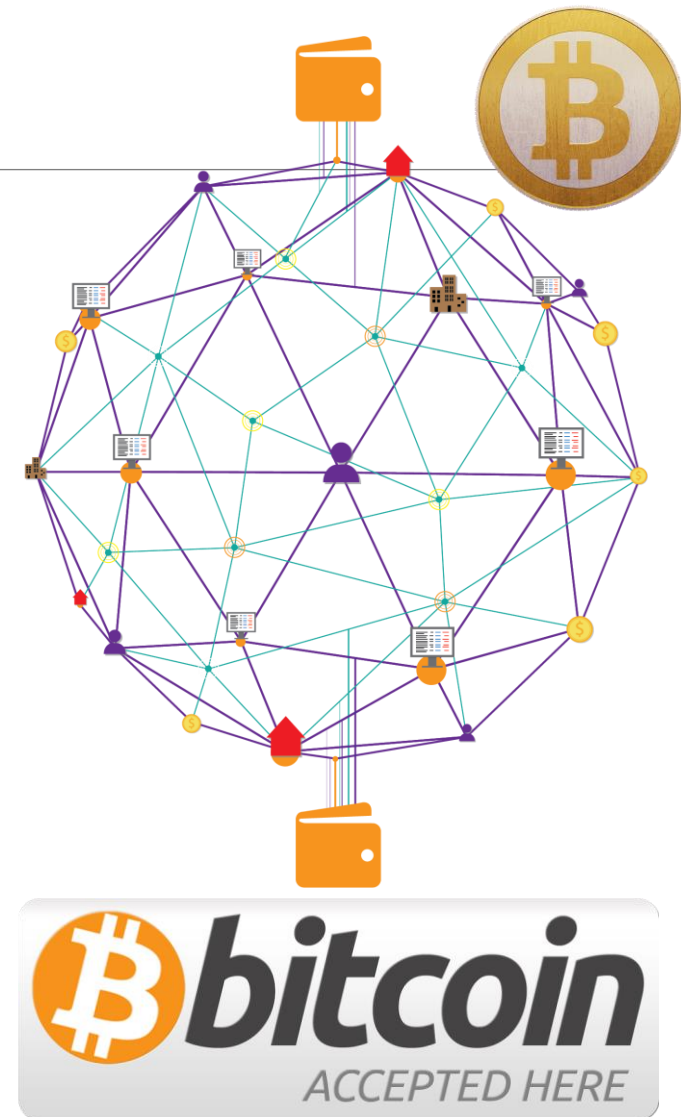
Why Bitcoin?

➤ Motivation

- Distrust of financial institutions
- Avoid the control of the traditional banking system

Any noncash transaction requires a trusted third-party administrator—commonly a bank or financial service provider.

The system forces participants to trust financial institutions that are not always trustworthy.

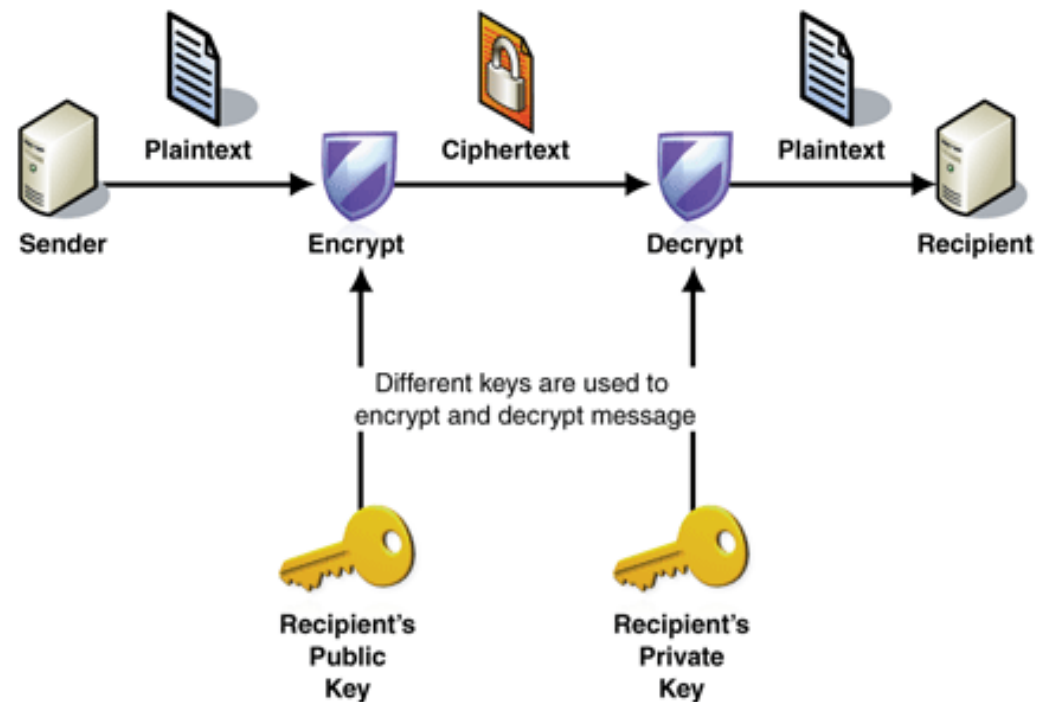
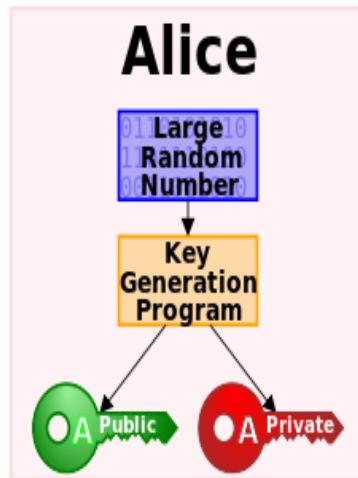


Cryptography in Bitcoin

Public-key Cryptography

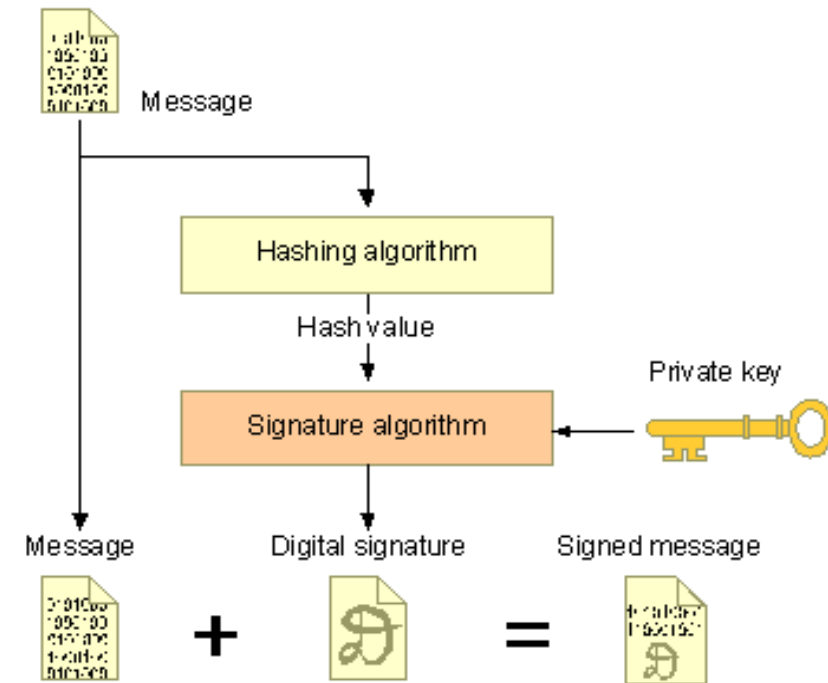
Key-pair: public-key and private-key

- In Bitcoin: Elliptic Curve *Secp256k1*



Digital Signature Algorithm

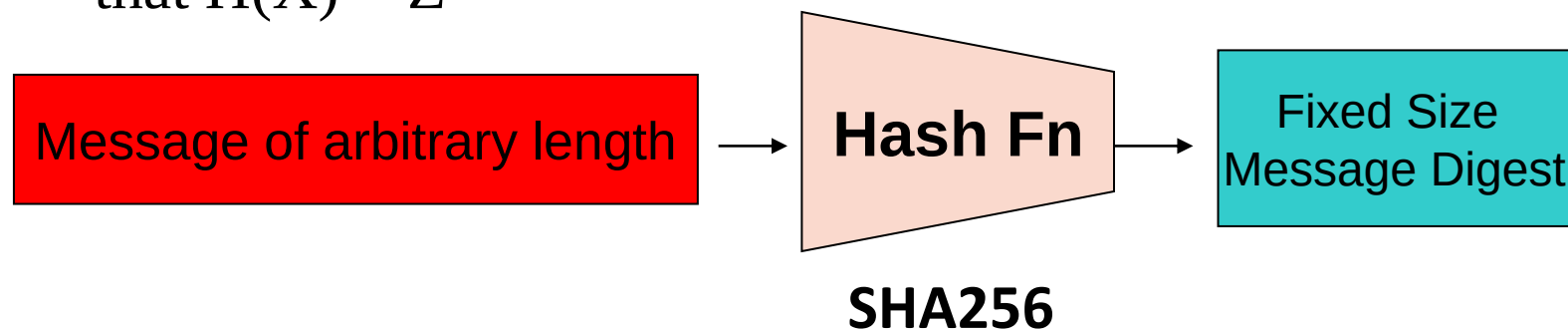
- First, create a message digest using a cryptographic hash
 - Then, encrypt the message digest with your private key
- In Bitcoin Elliptic Curve Digital Signature Algorithm (*ECDSA*, *Secp256k1*) with cryptographic hash (*SHA256*) is used for authentication/transaction verification



Authentication
Integrity
Non-repudiation

Cryptographic Hash Function

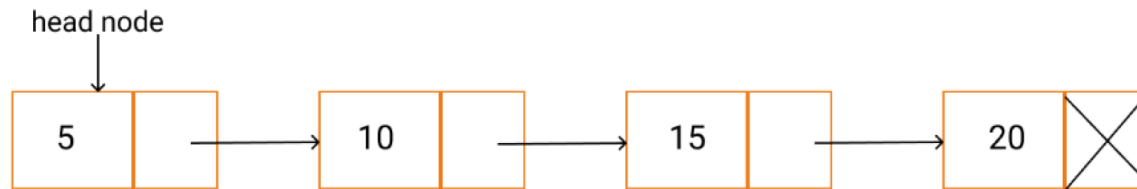
- Consistent: $\text{hash}(X)$ always yields same result
- Pre-image Resistance (One-way): any given Y in the output space, hard to find X s.t. $H(X) = Y$
- Second Pre-image Resistance: For a given message X , it is hard to find Y s.t. $X \neq Y$ and $H(X) = H(Y)$
- Collision resistance: given $H(W) = Z$, hard to find X such that $H(X) = Z$



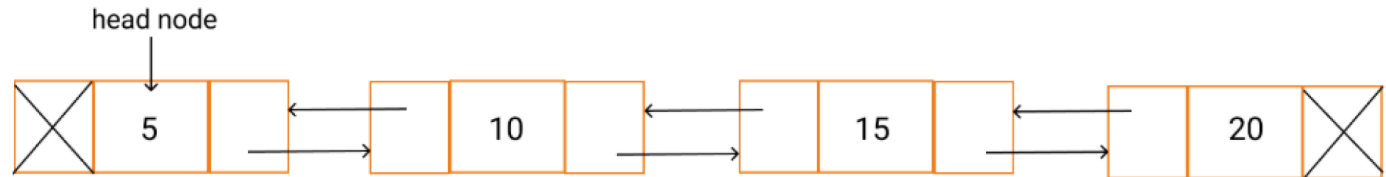
Linked List and Block Chains

- A linked list is a basic data structure where a series of data blocks are linked together.
- Each block contains the ID of the next one in the chain.
- It is possible to get to all of the blocks if we start with a pointer to the first block in the list.
- If each block also contains a hash of the previous block then it is called a Block Chain.
- It is not possible to insert a block in the middle of the list.
 - One of the hashes would disagree.

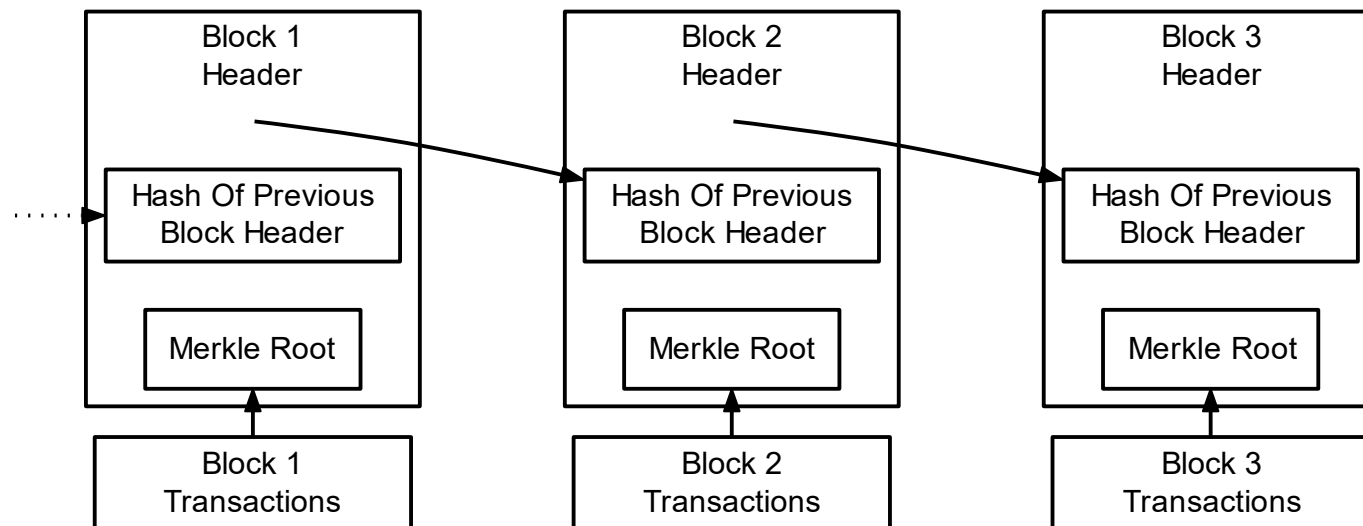
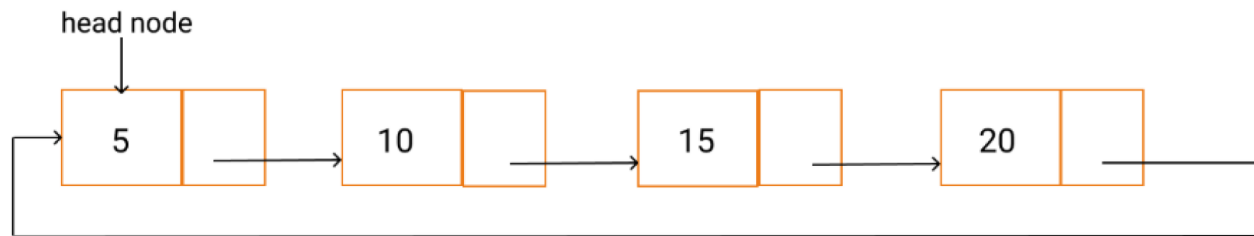
Singly Linked List



Doubly Linked List



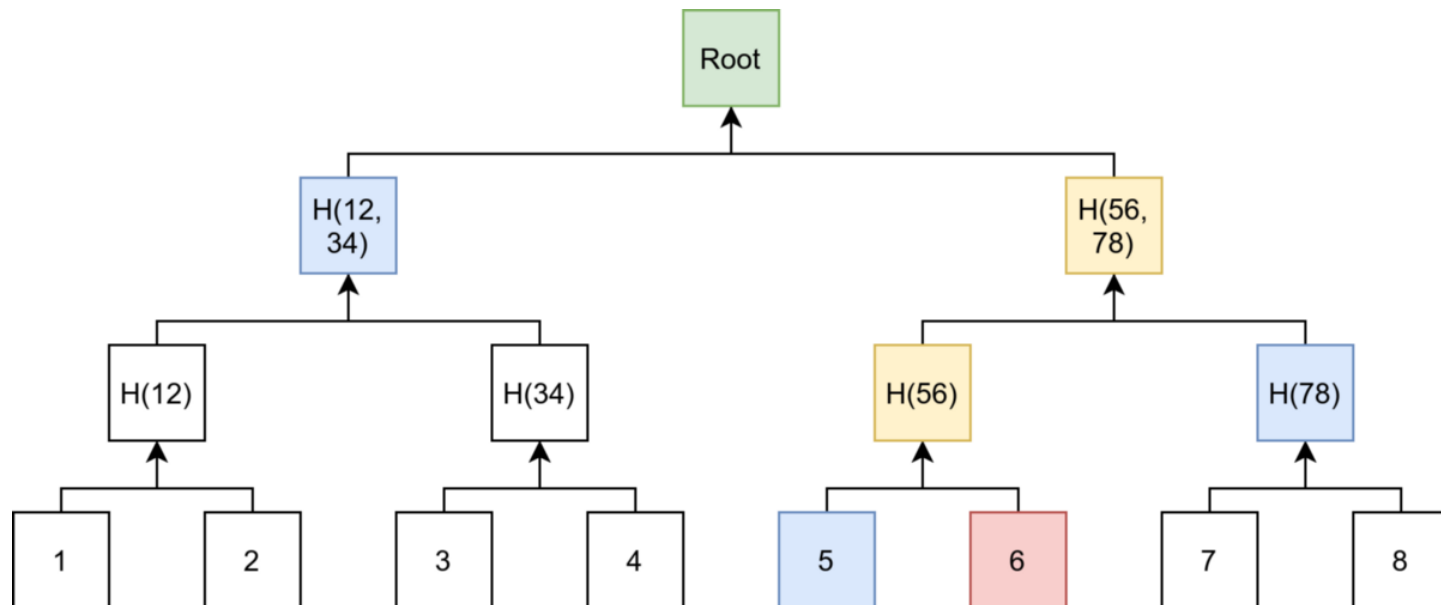
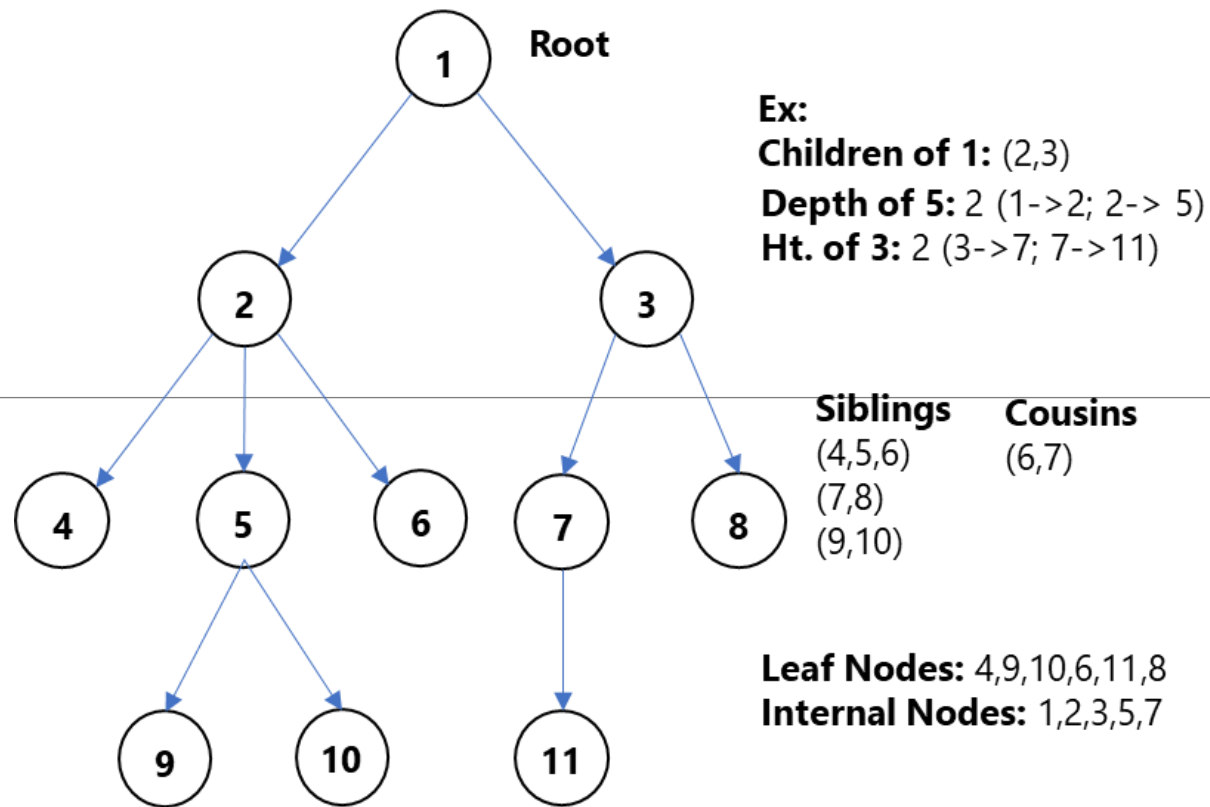
Circular Linked List



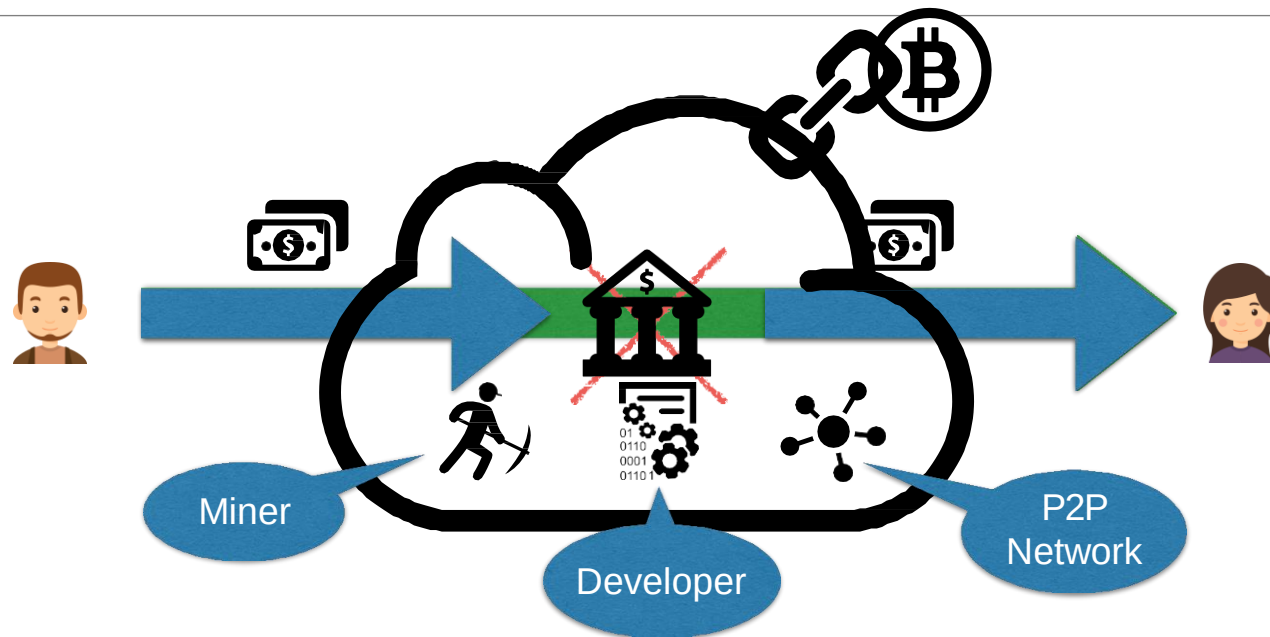
Simplified Bitcoin Block Chain

Data Structure: Binary and Merkle Tree

- A binary tree is like a linked list, but each block has pointers to two other blocks.
- In a sorted binary tree, the left and right linked blocks are in sorted order.
 - Left before, right after.
- It is possible to find a block, or prove that a block is not in a sorted tree, much quicker than a linked list.
 - Not all the blocks need to be checked.
- Each block in a binary tree can also contain the hashes of the two linked blocks. It is then called a Merkle tree.
 - It is impossible to insert a block in the middle of the tree afterwards.



Back to Bitcoin



1. How to perform secure **decentralized** payments in Bitcoin?
2. How to exchange **privacy-preserving** payments?
3. How to make decentralized systems **efficient**?



Summary

- ▶ A big picture of digital cash and Bitcoin as a cryptocurrency
 - ▶ Digital Cash, proposed by David Chaum
 - ▶ Bitcoin vs Traditional Banking System vs Digital Cash
- ▶ Remind of some cryptographic functions used in Bitcoin
 - ▶ Message Digest / Hash function
 - ▶ Public key cryptography with Digital Signature

Quiz

1. What is the difference between a peer-to-peer network and a centralised network? What things are harder to do with a peer-to-peer network?
2. Why do digital coins have a unique ID while pound coins do not? Is this similar to the unique ID on bank notes?
3. What is a cryptocurrency ledger and why is it important?
4. What is a blockchain and a Merkle tree? Why are they useful?